tf.ensure_shape Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/ensure_shape

> *Updates the shape of a tensor and checks at runtime that the shape holds.*

Function Signatures

```
tf.ensure_shape( x, shape, name=None )
```

Code Examples

```
tf.ensure_shape(
x, shape, name=None
)
```

```
tf.ensure_shape(
x, shape, name=None
)
```

```
x = tf.constant([[1, 2, 3],
[4, 5, 6]])
x = tf.ensure_shape(x, [2, 3])
```

```
x = tf.constant([[1, 2, 3],
```

```
[4, 5, 6]])
```

```
x = tf.ensure_shape(x, [2, 3])
```

```
x = tf.ensure_shape(x, [None, 3])
x = tf.ensure_shape(x, [2, None])
```

```
x = tf.ensure_shape(x, [None, 3])
```

Documentation

Updates the shape of a tensor and checks at runtime that the shape holds.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.ensure_shape

When executed, this operation asserts that the input tensor x's shape
is compatible with the shape argument.
See tf.TensorShape.is_compatible_with for details.

Use None for unknown dimensions:

If the tensor's shape is not compatible with the shape argument, an error
is raised:

During graph construction (typically tracing a tf.function),
tf.ensure_shape updates the static-shape of the result tensor by
merging the two shapes. See tf.TensorShape.merge_with for details.

This is most useful when you know a shape that can't be determined
statically by TensorFlow.

The following trivial tf.function prints the input tensor's
static-shape before and after ensure_shape is applied.

This lets you see the effect of tf.ensure_shape when the function is traced:

The above example raises tf.errors.InvalidArgumentError, because x's shape, (3,), is not compatible with the shape argument, (None, 3)

Inside a tf.function or v1.Graph context it checks both the buildtime and runtime shapes. This is stricter than tf.Tensor.set_shape which only checks the buildtime shape.

For example, of loading images of a known size:

When tracing a function, no ops are being executed, shapes may be unknown. See the Concrete Functions Guide
for details.

Returns

Raises